See instructions in attached doc.

https://gemini.google.com/app/9791da1f24f82439

User prompt: See instructions in attached doc.

Response: **Verdict:** Revision 4 is a major methodological step forward—we now agree with Fable on the $8.59/e^2$ continuous Brownian reflection constant, on generating fresh twin pairs independently per study market in Study 1, on restating the paper-account supermartingale on the world's filtration $\mathcal{G}_t$ (for fixed-cost binary contracts), and on requiring a measured thesis count from the existing ledger before reopening the reading-agent line. However, revision 4 still cannot be approved as written because **Study 2's confirmer design and Study 1's twin-market measure contain three critical breaks**: (1) in Study 2, requiring the posted isolated margin L on a hedged series to be at most $23.4\sigma_{\text{step}}$ is algebraically identical to requiring $\geq 200\%$ annualized hedged volatility on total posted margin, forcing $> 11\times - 22\times$ leverage that liquidates a dip-buying ladder on routine crypto dips (breaking vault fidelity), while placing the long and short legs on separate isolated margins creates a convex straddle payoff that breaks the Ville supermartingale under the null; (2) Study 2's fixed pre-freeze short $-\bar{\beta}_{\text{pre}} r_{m,t}$ fails to hedge a dip-buyer's regime-dependent inventory β_t, allowing pure post-freeze market drift $(\bar{\beta}_{\text{post}} - \bar{\beta}_{\text{pre}})\mu_m$ in either a bull or bear market to cross wealth 20 with $> 75\%$ probability under zero timing skill; and (3) in Study 1, adding a persistent AR(1) funding process ($\phi = 1 - 1/42$) into the planted coins' price returns induces strong positive price autocorrelation, allowing a funding-blind price-momentum program to capture $> 50\%$ of the reference effect on most twin pairs and trip Wall 3 before Session 1.

Ranked Findings (Part 1 of 2)

1. Critical — Study 2's $23.4\sigma_{\text{step}}$ isolated-margin bound and per-leg isolated margin conflict with vault fidelity, liquidate the dip-buyer on routine moves, and break the supermartingale null via straddle truncation

- **agreed:** Replacing the gate's fixed $L = 2$ (100σ) cap with a rule's own isolated-margin loss bound L is valid in principle if L is a single net position's worst-case loss and satisfies $L \leq \frac{2190}{S_{\text{planned}}}\sigma_{\text{step}}$.
- **disputed:**
  - **Where:** `architecture.md` §2.2 ("Kelly's fraction fits under `1/L` while a step's worst loss is at most `√2190 / S` standard deviations of a step: 23.4 at Sharpe 2, 33.4 at 1.4… A rule on isolated margin has one, the margin it posts for the step"), §6 ("The tested series. The replica's net return a step, minus a fixed short of its universe sized to its average exposure before the freeze, every leg on isolated margin posted at the step's start. The step's loss cannot exceed the margin posted, and that bound must be at most 23.4 standard deviations of a step"), and `arithmetic.py.txt` `confirmer()` lines 118–119.
  - **Construction:**
    1. **The $23.4\sigma_{\text{step}}$ bound requires $\geq 200\%$ annualized hedged volatility per dollar of total posted margin, destroying fidelity to a low-leverage dip-buyer:** Let $M_{\text{total}} = M_{\text{long}} + M_{\text{short}}$ be the isolated margin posted at the start of a 4-hour step across the replica's long ladder and the fixed short hedge. Because a 4-hour crypto candle can whipsaw (or long large-cap dips can diverge from the universe basket), the worst-case step loss is $L = M_{\text{total}}$ (or $\max(M_{\text{long}}, M_{\text{short}}) \geq M_{\text{total}}/2$ without whipsaws). Requiring $L \leq \frac{2190}{S}\sigma_{\text{step}} = 23.40\sigma_{\text{step}}$ at $S = 2$ is algebraically identical to requiring:

$$\frac{\sigma_{\text{ann, hedged}}}{M_{\text{total}}} = \frac{\sigma_{\text{step}}\ \ 2190}{L} \geq S = 2.0 \quad (200\%\ \text{annualized volatility on posted margin}).$$

Because the tested series $R_t^{\text{hedged}} = (\beta_t - \bar{\beta}_{\text{pre}})r_{m,t} + \epsilon_t$ strips out average market exposure $\bar{\beta}_{\text{pre}}$, its step volatility σ_{step} is only the **residual timing volatility**, whereas M_{long} and M_{short} must each margin the **full unhedged directional shock** of $\beta_t r_{m,t}$ and $-\bar{\beta}_{\text{pre}} r_{m,t}$. Specifically, for large-cap perps with 4-hour volatility $\sigma_{m,\text{step}} = 1.5\%$ (70.2% annualized), average exposure $\bar{\beta}_{\text{pre}} = 0.50$, and exposure SD $\text{SD}(\beta_t) = 0.25$, the hedged step SD is $\sigma_{\text{step}} = 0.25 \times 1.5\% = 0.375\%$. The bound $L \leq 23.4\sigma_{\text{step}}$ forces total posted margin to $M_{\text{total}} \leq 23.4 \times 0.375\% = 8.77\%$ ($M_{\text{long}} = M_{\text{short}} = 4.39\%$, i.e., $11.4\times$ to $22.8\times$ leverage). At 4.39% margin per leg, any 4.4% 4-hour dip liquidates the dip-buyer's filled long ladder instead of holding the dip, and any 8.8% 4-hour rally liquidates the $\bar{\beta}_{\text{pre}} = 0.50$ short hedge—failing the Fidelity wall against the actual cross-margined vault. Even worse, a dip-buying ladder places resting limit orders that require margin posted at step start; on non-dip steps where few orders fill ($\beta_t \approx 0$), $M_{\text{total}}/\sigma_{\text{step}}$ blows up far beyond 23.4. Conversely, posting realistic non-liquidating margin ($M_{\text{total}} = 50\%$) yields $L/\sigma_{\text{step}} = 133.3 > 23.4$, capping the bet at $1/L$ and collapsing 2-year power at Sharpe 2 back to the gate's capped regime.

2. **Posting "every leg on isolated margin" creates a convex straddle that breaks the Ville supermartingale under the null:** Section 2.2 notes that "clipping a loss raises the mean and breaks the null," yet §6 places "every leg on isolated margin posted at the step's start." When the long leg $\beta r_{m,t}$ and short hedge leg $-\beta r_{m,t}$ are placed on separate isolated margins $M_L = M_S = M$, any step move $|r_{m,t}| > M/\beta$ liquidates and truncates the losing leg at $-M$ while the winning leg earns uncapped upside:

$$R_t^{\text{hedged}} = \max(-M, \beta r_{m,t}) + \max(-M, -\beta r_{m,t}) = \begin{cases} 0 & \text{if } |\beta r_{m,t}| \leq M, \\ |\beta r_{m,t}| - M > 0 & \text{if } |\beta r_{m,t}| > M. \end{cases}$$

Under a zero-mean symmetric jump-diffusion null where $|\beta r_{m,t}| > M$ on 1% of steps by an average excess of 2%, separate per-leg isolated margins generate a positive drift of $+0.02\%$ per step ($+43.8\%$/year), violating $\mathbb{E}_0[R_t^{\text{hedged}} \mid \mathcal{F}_{t-1}] \leq 0$ and invalidating the 5% Ville bound.

- **Outcome:** Study 2 either fails its pre-freeze walls (the replica cannot simultaneously match the cross-margined vault's returns and satisfy $L \leq 23.4\sigma_{\text{step}}$ on recorded books) or invalidates its Type-I error bound via per-leg isolated-margin truncation.
- **Severity:** Critical.

2. Critical — Study 2's fixed pre-freeze short lets passive market drift be confirmed as edge in both bull and bear markets, and pre-freeze decomposition cannot detect it

- **agreed:** Stripping out passive market exposure is necessary so a long-only dip-buyer is not confirmed merely for holding positive crypto beta.
- **disputed:**
  - **Where:** `architecture.md` §6 ("Decomposition, with no waiting, descriptive only: how much of the reported Sharpe is the return of holding the vault's average exposure, and how much is timing… Claim, on rejection: from the freeze on, not every four-hour step of the frozen replica, hedged, had a non-positive expected net return given its past. The tested series. The replica's net return a step, minus a fixed short of its universe sized to its average exposure before the freeze") and §11.4 ("A wrong size

leaves market return in the tested series, and a rising market would then be confirmed as edge. The decomposition result is the check, and it is descriptive").

- **Construction:**

  1. **Step-level and window-level market return leakage under zero timing skill:** A dip-buying ladder has path-dependent exposure $\beta_t \in [0, 1]$: it accumulates inventory during drawdowns and unloads inventory in rallies. With a static short sized to pre-freeze average exposure $\bar{\beta}_{\text{pre}}$, the post-freeze step return is $R_t^{\text{hedged}} = (\beta_t - \bar{\beta}_{\text{pre}}) r_{m,t} - c_t$. Since β_t is determined by start-of-step inventory (and resting order fills), conditional on the past $\mathcal{F}_{t-1}$:

  $$\mathbb{E}[R_t^{\text{hedged}} \mid \mathcal{F}_{t-1}] = (\beta_t - \bar{\beta}_{\text{pre}}) \mathbb{E}[r_{m,t} \mid \mathcal{F}_{t-1}] - c_t.$$

  Even when the strategy has **zero timing skill** ($\text{Cov}(\beta_t, r_{m,t}) = 0$), any post-freeze shift in average ladder utilization $\bar{\beta}_{\text{post}} \neq \bar{\beta}_{\text{pre}}$ converts raw market drift μ_m directly into tested edge $(\bar{\beta}_{\text{post}} - \bar{\beta}_{\text{pre}}) \mu_m$:

     - *Bull-market false confirmation ($\bar{\beta}_{\text{post}} > \bar{\beta}_{\text{pre}}$, $\mu_m > 0$):* Suppose pre-freeze was a low-dip grind with $\bar{\beta}_{\text{pre}} = 0.25$, while post-freeze experiences deeper pullbacks that keep the ladder filled at $\bar{\beta}_{\text{post}} = 0.65$ during an $+80\%$/year bull market ($\mu_m = 0.80$, residual $\sigma_{\text{hedged}} = 0.16$). With zero timing skill, the residual beta $\Delta\bar{\beta} = +0.40$ earns $+32\%$/year (annualized Sharpe $S = 0.32/0.16 = 2.00$), crossing wealth 20 within two years with **77% probability**.

     - *Bear-market false confirmation ($\bar{\beta}_{\text{post}} < \bar{\beta}_{\text{pre}}$, $\mu_m < 0$):* Conversely, suppose pre-freeze had frequent dips ($\bar{\beta}_{\text{pre}} = 0.60$) and post-freeze enters a -60%/year bear market where the ladder stops out or sits lightly filled at $\bar{\beta}_{\text{post}} = 0.20$. Then $\Delta\bar{\beta} = -0.40$ (the tested series is net short the market by 40%), earning $(-0.40)(-0.60) = +24\%$/year from the market crash and confirming a passive short-market overhang as strategy skill.

  2. **Pre-freeze decomposition is tautologically blind to post-freeze beta mismatch:** By construction, $\bar{\beta}_{\text{pre}} = \frac{1}{T_{\text{pre}}} \sum_{t \leq 0} \beta_t$ makes the average residual exposure identically zero in the pre-freeze decomposition ($\bar{\beta}_{\text{pre}} - \bar{\beta}_{\text{pre}} = 0$). A pre-freeze descriptive check provides zero control over $\bar{\beta}_{\text{post}} - \bar{\beta}_{\text{pre}}$ after the freeze.

- **Outcome:** Allows Study 2's confirmer to reject the null at $W \geq 20$ purely from post-freeze market drift $\mu_m \neq 0$ whenever ladder utilization shifts between pre-freeze and post-freeze regimes.

- **Severity:** Critical.

3. Critical — Study 1's planted AR(1) funding process ($\phi = 1 - 1/42$) induces price momentum on the planted twin, allowing funding-blind momentum programs to beat half the reference and trip Wall 3

- **agreed:** Giving each study market m its own 32 fresh twin pairs (`m/fresh/0` to `m/fresh/31`) seeded from m's id completely resolves the Round 3 inter-market dependence bug across the 100 study markets.

- **disputed:**

  - **Where:** `architecture.md` §5.1 ("They differ in the planted coins' returns and, under N2, in those coins' variance paths, which feed on the planted return… A pick earns from the plant when, on at least 24 of its market's 32 pairs, its effect is at least half the locked reference program's effect on the same pair, and that effect is positive"), §5.5 Wall 3 ("A locked set of programs that cannot read funding (constant

long and short… momentum; reversal; a fixed random position at three leverages) is classified on 200 calibration markets each. At most 1% of those classifications earn from the plant. More means the score pays convexity on most pairs, and the measure is wrong"), §9 Finding 1, and `arithmetic.py.txt` line 188 (`phi = 1 - 1 / 42`).

- **Construction:**

  1. **Planting persistent AR(1) funding into price returns creates positive return autocorrelation that pays funding-blind price momentum:** Fable assumes a funding-blind program can only beat $0.5\Delta_{\text{ref}}$ via *convexity* ("More means the score pays convexity on most pairs"). In `gate/markets.py` (`arithmetic.py.txt` line 188), the funding signal s_t is a highly persistent AR(1) process with $\phi = 1 - 1/42 = 0.9762$ (long-run variance factor $\frac{1+\phi}{1-\phi} = 83$), whereas unplanted price returns $r_t(0)$ are serially uncorrelated noise (σ_r). Because the generator plants the edge by adding $+\alpha s_t$ to the planted coins' price returns $r_t(\alpha) = r_t(0) + \alpha s_t$, the planted coins' **price returns themselves become positively autocorrelated**:

     $$\mathrm{Cov}(r_t(\alpha), r_{t-k}(\alpha)) = \alpha^2 \mathrm{Cov}(s_t, s_{t-k}) = \phi^k \alpha^2 \sigma_s^2 > 0.$$

     Consider the funding-blind `momentum` program in Wall 3 that forms an exponential moving average of past price returns $M_t = (1-\lambda)\sum_{k=1}^{\infty} \lambda^{k-1} r_{t-k}(\alpha)$ with $\lambda = 0.95$ ($\sim$ 20-step lookback) and never reads `funding`. Averaging over 20 steps shrinks unplanted price-noise variance by $\frac{1-\lambda}{1+\lambda} = \frac{1}{39} = 0.0256$, while preserving $\frac{1-\lambda}{1+\lambda}\frac{1+\lambda\phi}{1-\lambda\phi} = 0.692$ of the persistent plant variance. At a modest per-step plant SNR of $\delta = \alpha\sigma_s/\sigma_r = 0.12$, the correlation between the funding-blind price EMA M_t and the current planted drift αs_t is:

     $$\rho(M_t, \alpha s_t) = \frac{\frac{(1-\lambda)\phi}{1-\lambda\phi}\delta}{\frac{1-\lambda}{1+\lambda} + 0.692\delta^2} = \frac{0.812 \times 0.12}{0.0256 + 0.0100} = 0.516 > 0.50.$$

     On the twin ($\alpha = 0$), $\rho = 0$. Thus, a funding-blind price-momentum program earns $> 50\%$ of the funding reference program's twin difference on a majority of pairs—either **tripping Wall 3 (**$> 1\%$ **pass rate) and halting Study 1 before Session 1**, or misclassifying ordinary price-momentum rules discovered by G100/the agent as "earning from the planted funding edge."

  2. **The 24-of-32 majority rule suffers a 78%–94% false-negative rate on genuine picks capturing 60%–70% of the reference edge:** As noted in §11.2, when the reference program sits near Wall 2's threshold (95% positive pairs, pair SNR $z_{\text{ref}} = 1.645$), a genuine funding pick capturing 70% of the reference effect with unit residual noise has single-pair success probability $q = \Phi(\frac{(0.7-0.5)\times 1.645}{1.25-0.70}) = 0.671$. Under `binom.sf(23, 32, q)`, a pick with $q = 0.671$ is rejected as failing **78.0% of the time** ($\mathbb{P}(\geq 24/32) = 0.220$), and at $q = 0.60$ (`arithmetic.py.txt` line 154) it is rejected **94.0% of the time** (`0.0596`).

- **Outcome:** Halts Study 1 at Wall 3 (or contaminates Study 1 with funding-blind price-momentum picks) and discards up to 78%–94% of partial funding discoveries.

- **Severity:** Critical.

CONTINUE FOR PART 2

User prompt: continue

Response: Ranked Findings (Part 2 of 2)

4. High — Study 2's pre-freeze Kelly fraction and power table ignore post-freeze volatility shifts ($\hat{\sigma}_{\text{pre}} \neq \sigma_{\text{post}}$), miscompute time-to-80%-power at Sharpe 1.4 as 4.4 years instead of 6.13 years, and overstate discrete heavy-tailed power via continuous Brownian formulas

- **agreed:** In continuous time for homoskedastic Gaussian returns with known volatility, the reflection formula gives $u_{0.80} = 2.9305$ ($N = 8.59/e^2$) and requires annualized Sharpe $S = 2.9305/\sqrt{\quad 2} = 2.072$ over two years when the rule's worst loss satisfies $L \leq 1/\lambda^*$.
- **disputed:**
  - **Where:** `architecture.md` §2.2 ("A bet sized for Sharpe 2 has power 0.77 within two years at a true Sharpe of 2, 0.47 at 1.4 and 0.90 at 2.4"), §2.3 Table ("Two years: 1.76 | 2.07 | 4.0"), §6 ("Test. Wealth starts at 1 and bets Kelly's fraction for a Sharpe of 2, computed from the replay before the freeze and registered… Power within one year and two: 0.12 and 0.28 at Sharpe 1; 0.22 and 0.47 at 1.4; 0.42 and 0.77 at 2; 0.57 and 0.90 at 2.4. Under no edge, 0.04. Eighty percent takes 1.5 years at 2.4 and 4.4 at 1.4"), and `arithmetic.py.txt` `series_power()` (lines 97–110) and `confirmer()` (lines 113–128).
  - **Construction:**
    1. **Freezing the unstandardized Kelly fraction from pre-freeze replay (`\hat{\sigma}_{\text{pre}}`) causes negative log-wealth drift when post-freeze volatility rises (`\sigma_{\text{post}} > \hat{\sigma}_{\text{pre}}`):** Section 6 registers the fixed betting fraction $\lambda_{\text{reg}} = \frac{2.0/\sqrt{\quad 2190}}{\hat{\sigma}_{\text{pre}}}$ from the pre-freeze replay. However, `series_power()` in `arithmetic.py.txt` (line 106) simulates `1 + bets * rng.normal(sharpe * dt, math.sqrt(dt), paths)`, tacitly assuming that post-freeze step volatility σ_{post} is identically equal to pre-freeze volatility $\hat{\sigma}_{\text{pre}}$. Let $r = \sigma_{\text{post}}/\hat{\sigma}_{\text{pre}}$ be the post-freeze to pre-freeze volatility ratio. When true post-freeze Sharpe is S_{post}, the effective standardized bet size is $b_{\text{eff}} = 2r$, giving annual log-wealth drift and diffusion spread:

       $$g(r) = 2rS_{\text{post}} - 2r^2, \qquad s(r) = 2r.$$

       If post-freeze crypto volatility is $1.5\times$ higher than pre-freeze ($r = 1.5$) while true Sharpe is $S_{\text{post}} = 1.4$ (the vault's recent figure), the strategy is over-bet beyond $2\times$ Kelly: annual log drift becomes **strictly negative** ($g(1.5) = 2(1.5)(1.4) - 2(1.5)^2 = -0.30$/year), cutting 2-year power from `0.47` to `0.34`. At $r = 2.0$, $g(2.0) = -2.40$/year and 2-year power at $S_{\text{post}} = 1.4$ collapses to `0.21` (and at $S_{\text{post}} = 2.0$, $g(2.0) = 0$ and 2-year power drops from `0.77` to `0.45`). Conversely, if post-freeze volatility is half of pre-freeze ($r = 0.5$), the strategy under-bets ($b_{\text{eff}} = 1.0$) and 2-year power drops from `0.77` to `0.58` at $S = 2.0$ and from `0.47` to `0.27` at $S = 1.4$.
    2. **Wrong formula for "4.4 years at 1.4" in §6:** Section 6 states that Study 2's test (which bets the fixed fraction $b = 2.0$ for Sharpe 2) takes "4.4 [years] at 1.4" to reach 80% power. Line 127 of `arithmetic.py.txt` computed `U80**2 / 1.4**2 = 4.38` years, which assumes the test bets **Kelly's fraction for $S = 1.4$ ($b = 1.4$)** rather than the registered bet $b = 2.0$. Under the registered bet $b = 2.0$ at true Sharpe $S = 1.4$, annual drift is $2(1.4) - 2 = 0.80$ and annual spread is 2.0: at $T = 4.38$ years, `crossing(0.80 * 4.38, 2 * sqrt(4.38))` is only **0.692 (69.2%)**. Reaching 80% power under the registered bet $b = 2.0$ actually takes 6.13 **years** (`crossing(0.80 * 6.13, 2 * sqrt(6.13)) = 0.800`), 40% longer than stated. (At $S = 2.4$ with $b = 2.0$, 80% power takes 1.52 years vs. 1.49 years.)

3. **Discrete-step and heavy-tail drag on the 2.07 threshold:** In `series_power()`, discrete 4-hour steps at $S = 2.072$ reach 20 in `0.786` of paths (requiring $S = 2.12$ for 80.0% discrete power), and under leptokurtic 4-hour perp returns ($\gamma_4 = 10$), fourth-moment log drag $-\frac{\gamma_4 - 1}{4}\lambda^4\sigma^4$ and discrete boundary overshoot reduce 2-year power at $S = 2.0$ by a further 3–5 points.

- **Outcome:** Overstates Study 2's robustness to volatility regime shifts between pre-freeze and post-freeze windows and understates the time required to reach 80% power at the vault's recent Sharpe of 1.4 (6.13 years vs. 4.4 years).

- **Severity:** High.

5. High — Study 2's pre-freeze fidelity check on a reverse-engineered vault conflates in-sample specification fitting with kernel validation and cannot verify sub-5-second passive ladder fills or isolated-margin step mechanics

- **agreed:** Replaying the kernel against a live Hyperliquid account's public history over archived L2 books is a valuable engineering test of fee, funding, and accounting plumbing that requires no forward waiting.

- **disputed:**

  - **Where:** `architecture.md` §6 ("Fidelity, with no waiting: the replica run by the kernel over the archive, against the vault's own public record over the same dates. This is the first check of the kernel's fills, fees and funding against an account that really traded, and it is a result about the machinery whatever the edge… Walls before any forward step: the archive's dates cover the vault's record; the replica reproduces the vault's daily returns within a tolerance registered before the comparison; the kernel can post isolated margin a step on recorded books").

  - **Construction:**

    1. **In-sample observational equivalence between reverse-engineered parameters and kernel fill errors:** Because the Astraeus vault's source code is not public ("a reverse-engineered vault: a long-only ladder that buys dips in large caps"), its ladder spacing, order sizes, and dip thresholds are inferred from the vault's historical trades and equity curve. Testing whether the reverse-engineered replica matches the vault's daily returns over *those same archive dates* tests the joint hypothesis of (reverse-engineered rule + kernel execution model). For example, if the kernel's fill model on 5-second S3 L2 snapshots overstates passive limit-order fill rates by 25% (by ignoring queue priority or missing sub-5-second wick reversals), a reverse-engineered ladder tuned to match the vault's historical daily returns simply widens its dip thresholds by 15–30 bps to match daily exposure, passing the daily-return tolerance wall while leaving the kernel's fill defect undetected. To test kernel fills independently of rule reverse-engineering, the kernel must replay the vault's **exact historical order/trade events** (from Hyperliquid's public vault trade history) or validate the reverse-engineered rule on a held-out pre-freeze sub-window not used to infer the ladder.

    2. **5-second S3 L2 snapshots miss sub-5-second liquidation wicks that drive dip-buying ladders:** Hyperliquid's S3 archive records L2 orderbook snapshots at 5-second intervals (`[hyperliquid-archive]`, §3). A limit-order dip ladder fills primarily during sub-second liquidation cascades that sweep book depth between 5-second snapshots and bounce back within 1–3 seconds. Replaying a limit-order ladder solely against 5-second L2 snapshots misses every intra-interval wick fill unless supplemented by tick/trade-stream extrema.

    3. **Mismatch between the cross-margined vault and Study 2's 4-hour isolated-margin step series:** The live vault runs continuously on a cross-margined account, whereas Study 2's confirmer tests a 4-hour stepped series with "every leg on isolated margin posted at the step's

start." If the fidelity wall compares the unhedged cross-margined replica against the vault, it never validates the isolated-margin step series actually fed to the confirmer; if it compares the 4-hour isolated-margin replica, step-boundary resets and isolated-margin liquidations break fidelity with the vault.

- **Outcome:** Prevents the daily-return fidelity check from separating kernel execution accuracy from in-sample reverse-engineering unless order-level fill replay and an out-of-sample pre-freeze holdout are required.
- **Severity:** High.

6. Medium — Study 1's 400-market bisection steps are too noisy for the $[0.13, 0.17]$ band, sessions killed by outages after evaluation 1 are still counted as misses, and N2 variance-path shifts alter twin scores

- **agreed:** Defining the null over the population of markets and the pinned `model_id`, halting on a `model_id` change, and voiding/rerunning sessions that exit non-zero with 0 evaluations (`spent == 0`) fix the deterministic-B_m, model-mixture, and 0-evaluation crash bugs from Round 3.
- **disputed:**
  - **Where:** `architecture.md` §5.1 ("They differ in the planted coins' returns and, under N2, in those coins' variance paths, which feed on the planted return"), §5.2 ("A session that carries another id is not counted; the study stops there… A session is void when the harness exits non-zero with no evaluation in the ledger… Every other session counts: one that ends by its own choice with no evaluation is a miss, and one cut off after an evaluation is judged on its pick"), §5.3 ("The plant's strength is found without a model, by bisection: the strength at which G100's rate on 2,000 calibration markets is between 0.13 and 0.17… At 0.3 the study is not powered… In all 20 of stage 3's null campaigns the agent stopped on its own after 6 to 22 evaluations, median 10.5"), §5.4 ("an agent at 0.3 against 0.15 stops here up to 8% of the time and ends undecided up to 36%"), §5.5 Wall 4 ("The bisection ends inside the band within eight steps"), and `arithmetic.py.txt` line 155 (`8 strengths of 400 markets and one of 2,000`).
  - **Construction:**
    1. **8-step bisection on 400-market samples has $\mathrm{SE} = 0.0179$, nearly as wide as the $[0.13, 0.17]$ target band:** While `arithmetic.py.txt` line 155 checks the final strength on 2,000 markets ($\mathrm{SE} = 0.0080$ at 0.15, $\mathrm{SE} = 0.0088$ at 0.19, so 0.19 is $2.28\ \mathrm{SE}$ above 0.17, not $2.5\ \mathrm{SE}$), the 8 bisection steps that *locate* that strength use only **400 markets per step** ($\mathrm{SE}_{400} = \quad 0.15 \times 0.85/400 = 0.0179$). Because a 95% noise fluctuation on 400 markets is ± 0.035—almost the full width (0.04) of the $[0.13, 0.17]$ target band—noisy bisection steps frequently bracket the wrong sub-interval, causing the 8th step to miss $[0.13, 0.17]$ on the 2,000-market check over **25% of the time** even when a smooth solution exists, falsely tripping Wall 4.
    2. **Under-powering and high futility risk when the agent stops early ($\mathrm{median}\ 10.5$ vs. G100's 100 evaluations):** Section 5.1 drops the equal-budget claim and pits the agent's self-stopped session ($\mathrm{median}\ 10.5$ evaluations in Stage 3) against G100's full 100 evaluations. If the agent's 10–20 evaluation search achieves a $2\times$ higher discovery rate than G100's 100 evaluations ($p_A = 0.30$ vs. $p_B = 0.15$–0.19), §5.3's table shows power is only `0.323` to `0.667` under least/independent overlap. Moreover, §5.4 states that an agent at 0.30 stops for futility "up to 8% of the time and ends undecided up to 36%" (quoting only $p_B = 0.15$), omitting `arithmetic.py.txt`'s output at the band edge $(0.30, 0.19)$ where futility stops hit **26.9%** and undecided hits **40.7%** (failing **67.7%** of the time).

3. **Mid-session operational crashes (`spent ⩾ 1`) are still counted as agent misses instead of resumed/voided:** Under §5.2, a session is voided *only* when it exits non-zero "with no evaluation in the ledger." Because the agent typically runs an initial baseline evaluation on turn 1 (`spent = 1`), any API timeout, provider 5xx drop, or sandbox crash occurring on turns 2–15 exits non-zero with `spent ⩾ 1`. Section 5.2 forces that crashed session to be "judged on its pick" (its turn-1 baseline evaluation, which fails), so a 5-session API instability streak after turn 1 still injects up to $5 \times (-0.5108) = -2.55$ into the SPRT log-likelihood ratio and triggers a false futility stop. Interrupted sessions exiting non-zero before the agent calls `finish` should be resumed from their last recorded turn on the same market rather than scored on a truncated turn-1 pick.

4. **N2 variance-path divergence between twins:** Under N2 (§5.1), adding planted returns $+\alpha s_t$ alters the GARCH conditional variance path $\sigma^2_{i,t}(\alpha) \neq \sigma^2_{i,t}(0)$ on the planted coins, which changes return volatility and margin liquidation timing on the planted twin even for strategies that do not trade funding.

- **Outcome:** Risks false Wall 4 failures during 400-market bisection, exposes early-stopping agent runs ($p_A = 0.30$) to a 26.9% false futility rate at the band edge, and penalizes mid-session API outages (`spent ⩾ 1`) as capability misses.

- **Severity:** Medium.

7. Low — Section 6's restated validity proof on the world's filtration holds for fixed-cost binary contracts, but applying it to §4's Hyperliquid perp theses requires fill-time indexing, isolated-margin gap bounds, and funding deduction

- **agreed:** Restating the paper-account supermartingale $V_t = C_t + \sum_{i \in U(t)} \frac{s_i}{c_i} \mathbb{E}[x_i \mid \mathcal{G}_t]$ on the world's filtration $\mathcal{G}_t$ (rather than the record's filtration $\mathcal{F}^{\text{rec}}_t$) completely resolves the Round 3 Doob-martingale and decision-predictability objections for fixed-cost binary payoffs $x_i \in [0, 1]$.

- **disputed:**

  - **Where:** `architecture.md` §4 ("A reading agent's bracketed theses on perps"), §6 last paragraph ("A purchase pays a stake the agent registers, at most 5% of cash, at the recorded all-in cost `c < 1`, and holds `stake/c` units. A recorded settlement `x` in [0, 1] pays `(stake/c)·x` into cash. The null is rejected when cash reaches 20. Null: for every purchase, `E[x | everything that has happened by its registration, recorded or not] ≤ c`"), and `arithmetic.py.txt` `account()` line 81.

  - **Construction:**

    1. **Measurability of recorded all-in cost c_j at registration time τ_j:** In §6's proof, a registration at τ_j lowers cash by s_j and adds $\frac{s_j}{c_j}\mathbb{E}[x_j \mid \mathcal{G}_{\tau_j}] \leq s_j$ to V_{τ_j}. This inequality requires the recorded all-in execution cost c_j to be known ($\mathcal{G}_{\tau_j}$-measurable) at the time τ_j in the conditioning set. If the agent registers a thesis at τ_j and its entry fill price c_j is determined at the next recorded Hyperliquid book snapshot $\tau_j^{\text{fill}} > \tau_j$, the null and supermartingale step must be indexed at τ_j^{fill} (when c_j is realized), or else random execution cost c_j creates a Jensen's inequality wedge $\mathbb{E}[x_j/c_j \mid \mathcal{G}_{\tau_j}] \neq \mathbb{E}[x_j \mid \mathcal{G}_{\tau_j}]/\mathbb{E}[c_j \mid \mathcal{G}_{\tau_j}]$.

    2. **Mapping §4's bracketed perp theses to $x \in [0, 1]$:** Because event-contract exchanges were dropped (§3), §4's reading agent trades **bracketed theses on Hyperliquid perps**. For a multi-day perp bracket to satisfy $x \in [0, 1]$ and $\mathbb{E}_0[x \mid \mathcal{G}_{\tau_j}] \leq c$ on recorded books, the account specification must enforce (a) per-position isolated margin equal to `stake` so a price gap past the stop-loss bracket cannot produce $x < 0$ (losses exceeding `stake`), and (b) deduction of

cumulative hourly funding payments inside the bracket's payoff x (otherwise a perp with positive funding carry has $\mathbb{E}[x_{\text{price}} \mid \mathcal{G}_\tau] \neq c$).

3. **Terminal pending-payout bug in `account()` (line 81):** Section 6 states that "the null is rejected when cash reaches 20", but `arithmetic.py.txt` line 81 evaluates `reached ⊨ cash + pending.sum(axis=1) ⩾ 20` at the end of week 104, counting unrecorded `pending` positions as realized `cash` at the horizon (at `bets = 33, hold = 5, edge = 0`, line 81 inflates the null crossing rate from `0.0235` to `0.0242`).

- **Outcome:** Leaves minor specification gaps between §6's binary-contract formula (`c < 1`, `x in [0, 1]`) and §4's Hyperliquid perp bracket design, plus a one-line horizon mismatch in `account()`.
- **Severity:** Low.

Convergence Ledger: Agreed Points vs. Open Disputes

Points Now Agreed

1. **Continuous stopping constant ($8.59/e^2$ and Sharpe 2.07):** We agree with Fable's continuous Brownian reflection derivation ($u_{0.80} = 2.9305$, $u_{0.80}^2 = 8.59$, and $2.9305/\sqrt{2} = 2.07$ over two years when Kelly is uncapped).
2. **Reading-agent threshold and parking the thesis track:** We agree that at a net edge ceiling of $e = 0.05$, reaching 80% power within two years requires ≥ 33 independent bets/week settled daily, and that the reading-agent track should stay parked until the advisor counts registered theses/week in the existing thesis ledger.
3. **Q2 paired-arm discordance scaling:** We agree with Fable's §4 analysis that pairing an agent with and without text filings cancels market variance where the two arms agree, so power scales with the count of *discordant* forecasts.
4. **Study 1 per-market fresh twin pairs (`m/fresh/0..31`):** Generating 32 fresh twin pairs per study market from `m`'s own seed completely fixes the Round 3 inter-market dependence bug across the 100 study markets.
5. **Study 1 population null and `model_id` pinning:** Stating the null over the population of markets and the pinned `model_id` (and stopping if `model_id` changes) resolves the deterministic-B_m and model-mixture bugs.
6. **Study 1 voiding of 0-evaluation crashes:** Voiding and rerunning sessions that exit non-zero with 0 evaluations (`spent == 0`) fixes pre-evaluation startup crashes.
7. **Section 6's validity proof on the world's filtration $\mathcal{G}_t$:** Restating the supermartingale V_t on $\mathcal{G}_t$ resolves our Round 3 filtration and outcome-dependent delay critique for fixed-cost binary contracts.
8. **Paper Findings 7, 10, and 11 (`argon-paper-findings.md`):** We agree with Fable's dispositions of Paper Findings 11 (entering the stationary bootstrap variance understatement into the defect table at funding shares 0.05–0.25), 10 (quoting the suite's tolerance as the certified size), and 7 (noting the algebraic bound $\hat{\gamma}_4 \geq \hat{\gamma}_3^2 + 1$ on sample moments and the 91-day wall).

Disputes Still Open

#	Severity	Section	Open Dispute	Required Fix to Converge
1	**Critical**	§2.2, §6	**Study 2's $23.4\sigma_{\text{step}}$ bound and per-leg isolated margin:** $L \leq 23.4\sigma_{\text{step}}$ requires $\geq 200\%$ annualized hedged vol on total posted margin, forcing $> 11\times$–$22\times$ leverage that liquidates a dip-buyer on 4.4%–8.8% moves (breaking vault fidelity), while separate per-leg isolated margins create a convex straddle that violates $\mathbb{E}_0[R_t^{\text{hedged}}] \leq 0$.	Margin the replica and hedge inside a single cross-margined sub-account (or net portfolio stop-loss at $-L_{\text{net}}$) rather than per-leg isolated margins, and verify pre-freeze whether a non-liquidating L_{net} achieves $\leq 23.4\sigma_{\text{step}}$; if not, size the bet at $\min(\lambda_{\text{reg}}, 1/L_{\text{net}})$ and report the resulting capped power.
2	**Critical**	§6, §11.4	**Study 2's fixed pre-freeze short hedge:** Because a dip-buyer's exposure β_t is regime-dependent, a static pre-freeze short $-\bar{\beta}_{\text{pre}} r_{m,t}$ leaves $(\beta_t - \bar{\beta}_{\text{pre}}) r_{m,t}$ unhedged, allowing bull or bear market drift $(\bar{\beta}_{\text{post}} - \bar{\beta}_{\text{pre}})\mu_m$ to cross $W = 20$ with $> 75\%$ probability under zero timing skill.	Either (a) dynamically hedge start-of-step inventory $-\beta_{t-1} r_{m,t}$ (testing high-frequency intra-step fill timing) or (b) state explicitly in Study 2's claim that rejection confirms *joint dip-accumulation + market drift relative to fixed* $\bar{\beta}_{\text{pre}}$, not market-neutral timing skill.
3	**Critical**	§5.1, §5.5	**Study 1's AR(1) plant pays funding-blind price momentum & 24/32 rule drops partial picks:** Planting persistent AR(1) funding ($\phi = 1 - 1/42$) into returns induces price autocorrelation ($\rho = 0.516 > 0.50$), letting Wall 3's funding-blind `momentum` program pass on a majority of pairs; meanwhile 24/32 rejects $q = 0.67$ picks 78% of the time.	Demean or randomize the planted funding sign at a faster frequency (or orthogonalize planted returns against lagged price EMA) so funding-blind price momentum has zero expected edge, and increase pair count from 32 to 64 (or lower threshold from $0.5\Delta_{\text{ref}}$ to $0.35\Delta_{\text{ref}}$) so $0.7\Delta_{\text{ref}}$ picks pass $> 85\%$ of the time.
4	**High**	§2.2, §6, `arithmetic.py`	**Study 2's pre-freeze Kelly sensitivity and 4.4-year error:** Freezing λ_{reg} on $\hat{\sigma}_{\text{pre}}$ yields negative log drift ($g = -0.30$) at $S = 1.4$ if $\sigma_{\text{post}} = 1.5\hat{\sigma}_{\text{pre}}$; and 80% power at $S = 1.4$ under Study 2's registered bet $b = 2$ takes **6.13 years**, not 4.4 years (`U80^2/1.4^2` assumes betting $b = 1.4$).	Scale each step's bet by trailing pre-step realized variance $\hat{\sigma}^2_{t-1}$ (capped at $1/L_t$, which remains $\mathcal{F}_{t-1}$-predictable), and fix line 127 of `arithmetic.py.txt` and §6 to report 6.13 years at $S = 1.4$ when betting for Sharpe 2.

#	Severity	Section	Open Dispute	Required Fix to Converge
5	**High**	§6	**Study 2's in-sample fidelity wall on 5s L2 snapshots:** Matching daily returns of a reverse-engineered ladder on the same archive dates used to infer the ladder cannot separate kernel fill bugs from parameter tuning, and 5s L2 snapshots miss sub-5s liquidation wicks.	Split the pre-freeze archive into an inference window and an out-of-sample fidelity holdout, replay the vault's exact public fills to test kernel accounting directly, and incorporate trade-stream high/lows between 5s L2 snapshots.
6	**Medium**	§5.2–5.5	**Study 1's 400-market bisection noise, `spent ⩾ 1` crash rule, and $(0.3, 0.19)$ futility rate:** 400-market bisection steps ($\mathrm{SE} = 0.0179$) fail the $[0.13, 0.17]$ check $> 25\%$ of the time; outages killing a session after turn 1 (`spent ⩾ 1`) are still scored as misses; and $(0.3, 0.19)$ has a 26.9% futility rate.	Use $\geq 1,000$ markets in the final 3 bisection steps (or stochastic approximation), resume sessions interrupted by non-zero harness exits from their last checkpointed turn, and quote the 26.9% futility / 40.7% undecided figures for $(0.3, 0.19)$ in §5.4.
7	**Low**	§4, §6, `arithmetic.py`	**Perp bracket mapping in §6's account & line 81 of `account()`:** Applying §6's binary formula (`c < 1, x in [0, 1]`) to §4's perp brackets requires indexing at fill time τ^{fill}, isolated margin for gap risk, and deducting funding inside x; line 81 of `account()` counts `pending` as `cash`.	Index the null at fill time τ^{fill}, define $x \in [0, 1]$ net of funding on isolated margin for perp brackets, and remove `+ pending.sum(axis=1)` from line 81 of `arithmetic.py.txt`.